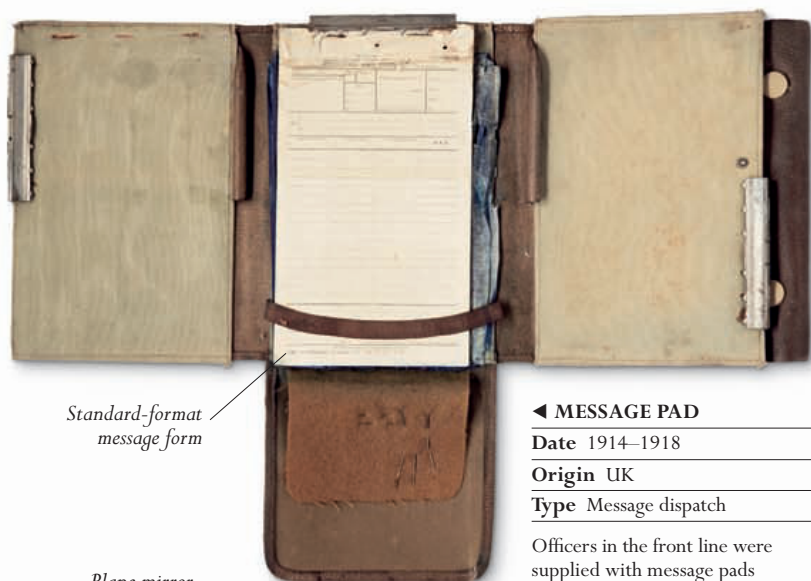


COMMUNICATIONS EQUIPMENT OF WORLD WAR I

As battlefields and armies grew in size, the need for effective long-distance communication became more urgent. Paper messages were carried by human couriers, as well as carrier pigeons and dogs, with varying degrees of reliability. Traditional signalling systems such as semaphore, flares, and heliographs were joined on the battlefield by new forms of telegraphy, based on electricity. Cable telegraphy, which depended on long wires connecting fragile telegraph machines, and wireless telegraphy, which demanded cumbersome hardware, were vulnerable to interception by the enemy. Notably, the entry of America into the conflict in 1917 was precipitated by the interception of a German telegram.



Standard-format message form

◀ MESSAGE PAD

Date 1914–1918

Origin UK

Type Message dispatch

Officers in the front line were supplied with message pads bound in leather and canvas covers. Carbon paper was used to produce a duplicate copy, which the sender retained.



Plane mirror

◀ HELIOGRAPH MARK V

Date 1916

Origin UK

Type Signalling

The first practical heliograph – a mirror used for flashing sunlight – was developed in India in the 1860s. Lightweight, needing no power source, and difficult to intercept, they were extremely useful and remained in service until the 1970s.



Flags were made from serge or silk – silk flags were quicker to manipulate

◀ SEMAPHORE FLAGS

Date 1914–1918

Origin UK

Type Signalling

The semaphore flag signalling system was developed for use as a way of communicating between ships in the 19th century, and remained in use until rendered obsolete by the introduction of lightweight radios. A practised signaller could signal a dozen or more words per minute.



Leather collar with secure pocket

◀ DOG COLLAR

Date 1914–1918

Origin UK

Type Message dispatch

Under some circumstances, usually in areas where it was deemed too dangerous to send a man, trained dogs were used to carry messages, secreted either in harnesses or simple collars.

▼ WHEATSTONE TELEGRAPH

Date 1914–1918

Origin UK

Type Telegraphy

Based on technology developed by British scientist Charles Wheatstone in the 1830s, the Wheatstone telegraph used a punch-tape system to transmit up to 100 words per minute.



Reader for punched paper tape